Free-space Continuous-variable Quantum Key Distribution

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Outline

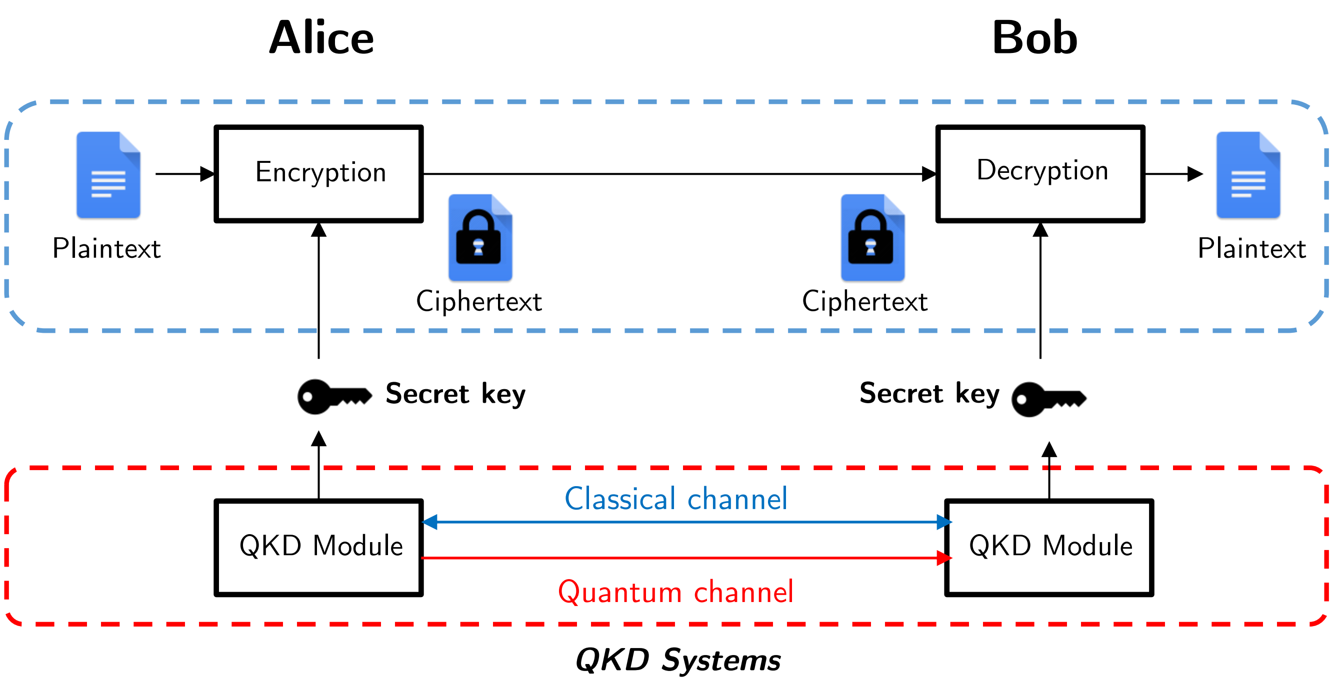
I. Review of Continuous-variable Quantum Key Distribution

II. My Focus: Free-space CV-QKD

Quantum Key Distribution

- Recently, quantum computers have been advancing rapidly ⇒ *It poses a growing threat to key distribution systems using public-key cryptography (PKC).*
  - *PKC-based key distribution*: Relies on the hardness of solving certain mathematical problems
- To address this problem, a solution is **quantum key distribution (QKD)**
  - *QKD*: Relies on the law of quantum mechanics to generate secret keys

=> The shared key can be secure given the unlimited computational power of adversaries.



QKD Classification

- QKD can be classified into three main categories based on how they encode information: (1) discrete-variable (DV), (2) coherent continuous-variable (CV), and (3) non-coherent CV-QKD

	DV-QKD	Non-coherent CV-QKD	Coherent CV-QKD
Encoding	Discrete quantum state (e.g., photon polarization)	Intensity	Phase and amplitude of coherent state
Detectors	Single-photon detectors	Direct detectors	Coherent detectors
Conventional devices compatibility	No (Require single-photon sources/detectors)	Yes	Yes
Security proof robustness	High	Limited	Normal

Focus of this presentation

CVQKD: GG02 Protocol

- **The GG02 protocol** is one of the well-known coherent CV-QKD protocols.

- **Considered scenario**:

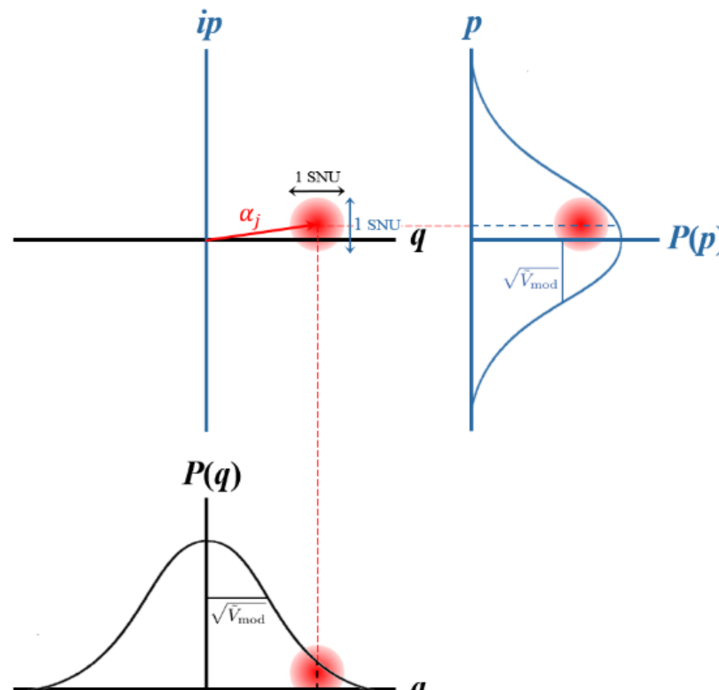
Alice and Bob want to share a secret key for confidential communication

A passive eavesdropper (Eve) tries to obtain the information of the key.

- **Step 1:** Alice generates two random numbers, q and p, sampled independently from a zero-mean normal distribution $\mathcal{N}(0, \tilde{V}_{\text{mod}})$

- **Step 2:** Alice prepare a coherent state $|q + ip\rangle$ and transmit to Bob.

$\tilde{V}_{\text{mod}}$: modulation variance



GG02 Protocol: Step-by step (cont.)

- **Step 3**: Upon receiving a coherent state, Bob randomly selects a quadrature to measure.
- **Step 4**: Bob informs Alice regarding the quadratures he selected via a public channel. Alice keeps the corresponding values and discards the remaining.
- **Step 5**: Both sides conduct the post-processing phase to generate the secret key.

Alice

q	0.9	2.4	1.3	2.1	0.5
p	1.2	0.9	5.1	2.7	1.5
Coherent state	$\|0.9+1.2i\rangle$	$\|2.4+0.9i\rangle$	$\|1.3+5.1i\rangle$	$\|2.1+2.7i\rangle$	$\|0.5+1.5i\rangle$

Bob

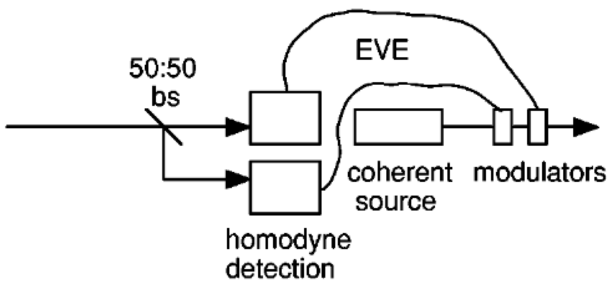
Selected quadrature	p	p	q	p	q
Detected output	**1.25**	**0.88**	**1.21**	**2.9**	**0.45**

The Security of GG02 Protocol

- Let's consider the presence of Eve
- Eve cannot copy the quantum states due to *the no-cloning theorem.*

=>Thus, Eve needs to directly measure the quantum state to get the information.

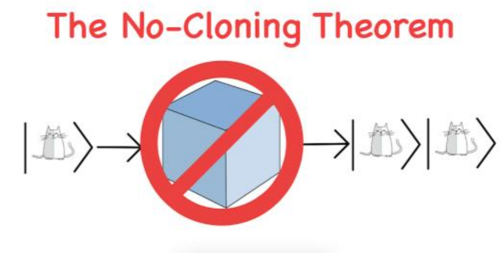
- Eve's strategy: splits the beam in half, measures both quadratures, create another coherent states, and sends them to Bob



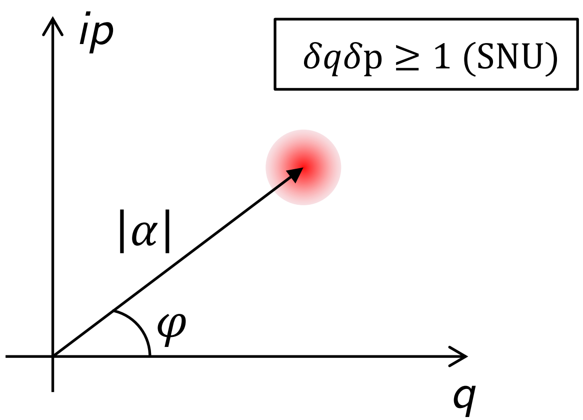
⇒ Eve can not measure the exact value of the coherent state

⇒ Eve will introduce excess noise to the system, which is used to analyze Eve's knowledge of the key.

(1) *No-cloning theorem:* It is impossible to create a perfect copy of an unknown quantum state.

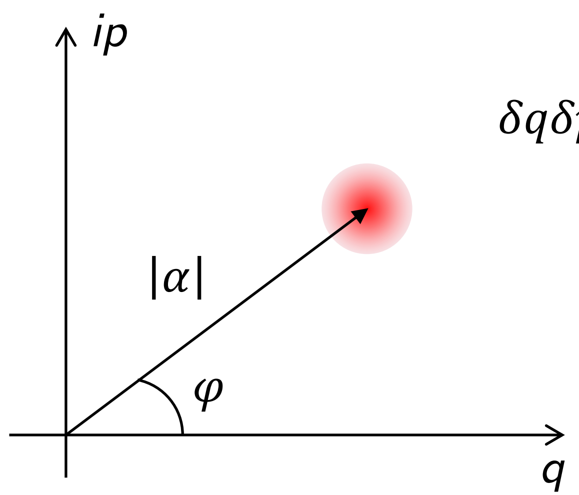


(2) Heisenberg uncertainty principle for field quadratures: The uncertainties in the quadratures of light cannot be simultaneously zero



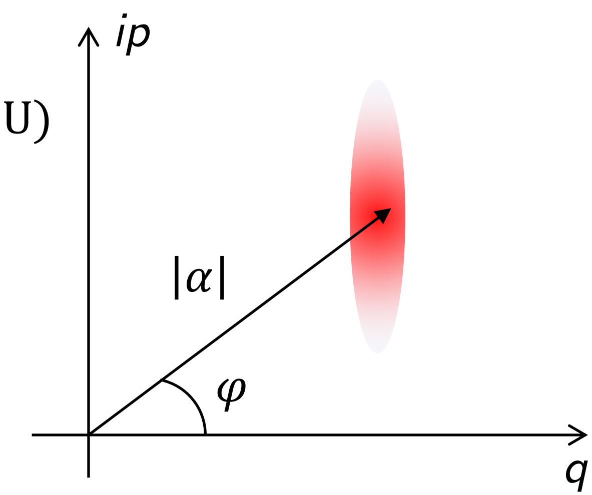
CV-QKD Variants: Squeezed States

- Besides GG02, there are several variants of CV-QKD protocols.
- One variant relies on the use of squeezed states.
- **A squeezed state** is a quantum state where the uncertainty in one quadrature is reduced below the vacuum noise .
- Pros: Can reduce the uncertainty when measuring a specific quadrature => Improve the system performance in some highly noisy channels
- Cons: Require a sophisticated laser source



Coherent state

$$\delta q \delta p \geq 1 \text{ (SNU)}$$



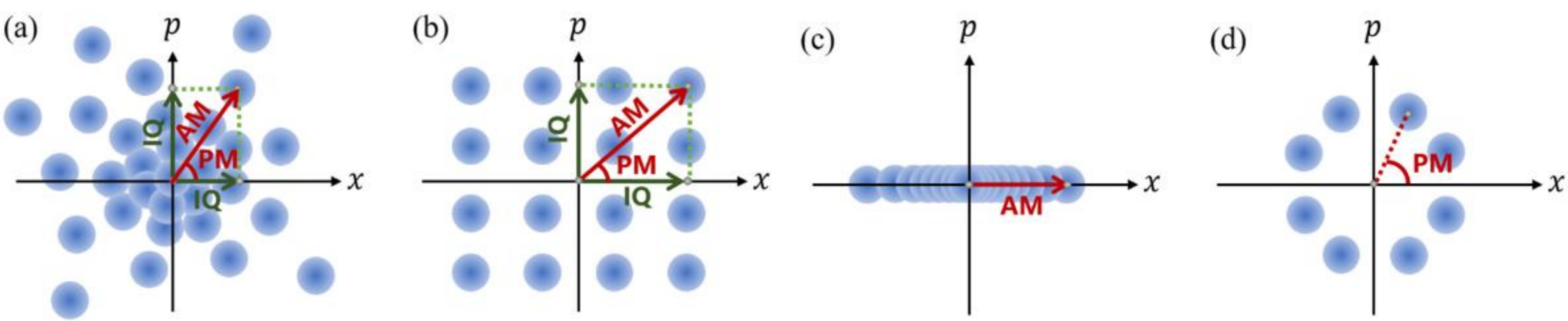
Squeezed state

CV-QKD Variants: Discrete Modulation

- *Practical implementations of GM-CV-QKD face several difficulties due to hardware limitations.*
  - Limited peak power of transmitters
  - Finite resolution of converters

$\Rightarrow$ Discrete-modulated (DM) CV-QKD

- **Key idea**: Alice randomly select a coherent state from a **predefined set.**
- Disadvantage: Less mature security proof



Gaussian-modulated CV-QKD (GG02) | Quadrature-amplitude-modulated CV-QKD | Amplitude-modulated CV-QKD | Phase-modulated CV-QKD

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